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Spectrum sensing during initial access to a full-duplex cell

Abstract

Apparatus, methods, and computer program products for using spectrum sensing and supporting FD operations are provided. An example method may include performing a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. The example method may further include selecting a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to communication systems, and more particularly, to wireless communication systems with full-duplex (FD) cells.

INTRODUCTION

(2) Wireless communication systems are widely deployed to provide various telecommunication services such as telephony, video, data, messaging, and broadcasts. Typical wireless communication systems may employ multiple-access technologies capable of supporting

communication with multiple users by sharing available system resources. Examples of such multiple-access technologies include code division multiple access (CDMA) systems, time division multiple access (TDMA) systems, frequency division multiple access (FDMA) systems, orthogonal frequency division multiple access (OFDMA) systems, single-carrier frequency division multiple access (SC-FDMA) systems, and time division synchronous code division multiple access (TD-SCDMA) systems.

(3) These multiple access technologies have been adopted in various telecommunication standards to provide a common protocol that enables different wireless devices to communicate on a municipal, national, regional, and even global level. An example telecommunication standard is 5G New Radio (NR). 5G NR is part of a continuous mobile broadband evolution promulgated by Third Generation Partnership Project (3GPP) to meet new requirements associated with latency, reliability, security, scalability (e.g., with Internet of Things (IoT)), and other requirements. 5G NR includes services associated with enhanced mobile broadband (eMBB), massive machine type communications (mMTC), and ultra-reliable low latency communications (URLLC). Some aspects of 5G NR may be based on the 4G Long Term Evolution (LTE) standard. There exists a need for further improvements in 5G NR technology. These improvements may also be applicable to other multi-access technologies and the telecommunication standards that employ these technologies.

BRIEF SUMMARY

(4) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(5) In an aspect of the disclosure, a method, a computer-readable medium, and an apparatus at a user equipment (UE) are provided. The apparatus may include a memory and at least one processor coupled to the memory. The memory and the at least one processor coupled to the memory may be further configured to perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or cross link interference (CLI) associated with one or more cells of a base station during initial access. The memory and the at least one processor coupled to the memory may be further configured to select a random access channel (RACH) occasion (RO) or a cell of the one or more cells based on the acquired information about the FD operation or CLI.

(6) In another aspect of the disclosure, a method, a computer-readable medium, and an apparatus are provided. The apparatus may include a memory and at least one processor coupled to the memory. The memory and the at least one processor coupled to the memory may be configured to transmit, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access. The memory and the at least one processor coupled to the memory may be further configured to receive, from the UE, communication based on a selection of a RO or a cell of the one or more cells based on the acquired FD operation or CLI.

(7) To the accomplishment of the foregoing and related ends, the one or more aspects comprise the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network.
(2) FIG. **2**A is a diagram illustrating an example of a first frame, in accordance with various aspects of the present disclosure.
(3) FIG. **2**B is a diagram illustrating an example of DL channels within a subframe, in accordance with various aspects of the present disclosure.
(4) FIG. **2**C is a diagram illustrating an example of a second frame, in accordance with various aspects of the present disclosure.
(5) FIG. **2**D is a diagram illustrating an example of UL channels within a subframe, in accordance with various aspects of the present disclosure.
(6) FIG. **3** is a diagram illustrating an example of a base station and user equipment (UE) in an access network.
(7) FIG. **4**A illustrates an example type of FD communication.
(8) FIG. **4**B illustrates an example type of FD communication.
(9) FIG. **4**C illustrates an example type of FD communication.
(10) FIG. **5**A illustrates an example use case of FD communication.
(11) FIG. **5**B illustrates an example use case of FD communication.
(12) FIG. **5**C illustrates an example use case of FD communication.
(13) FIG. **6** illustrates example communications between base station and UE(s) in a wireless communication system.
(14) FIG. **7**A illustrates an example sensed spectrum.
(15) FIG. **7**B illustrates an example sensed spectrum.
(16) FIG. **7**C illustrates an example sensed spectrum.
(17) FIG. **8** is a flowchart of a method of wireless communication.
(18) FIG. **9** is a flowchart of a method of wireless communication.
(19) FIG. **10** is a flowchart of a method of wireless communication.
(20) FIG. **11** is a diagram illustrating an example of a hardware implementation for an example apparatus.
(21) FIG. **12** is a diagram illustrating an example of a hardware implementation for an example apparatus.

DETAILED DESCRIPTION

(22) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(23) Several aspects of telecommunication systems will now be presented with reference to various apparatus and methods. These apparatus and methods will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(24) By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central

processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software shall be construed broadly to mean instructions, instruction sets, code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, etc., whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise.

(25) Accordingly, in one or more example embodiments, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media can comprise a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

(26) While aspects and implementations are described in this application by illustration to some examples, those skilled in the art will understand that additional implementations and use cases may come about in many different arrangements and scenarios. Innovations described herein may be implemented across many differing platform types, devices, systems, shapes, sizes, and packaging arrangements. For example, implementations and/or uses may come about via integrated chip implementations and other non-module-component based devices (e.g., end-user devices, vehicles, communication devices, computing devices, industrial equipment, retail/purchasing devices, medical devices, artificial intelligence (AI)-enabled devices, etc.). While some examples may or may not be specifically directed to use cases or applications, a wide assortment of applicability of described innovations may occur. Implementations may range a spectrum from chip-level or modular components to non-modular, non-chip-level implementations and further to aggregate, distributed, or original equipment manufacturer (OEM) devices or systems incorporating one or more aspects of the described innovations. In some practical settings, devices incorporating described aspects and features may also include additional components and features for implementation and practice of claimed and described aspect. For example, transmission and reception of wireless signals necessarily includes a number of components for analog and digital purposes (e.g., hardware components including antenna, RF-chains, power amplifiers, modulators, buffer, processor(s), interleaver, adders/summers, etc.). It is intended that innovations described herein may be practiced in a wide variety of devices, chip-level components, systems, distributed arrangements, aggregated or disaggregated components, end-user devices, etc. of varying sizes, shapes, and constitution.

(27) FIG. **1** is a diagram illustrating an example of a wireless communications system and an access network **100**. The wireless communications system (also referred to as a wireless wide area network (WWAN)) includes base stations **102**, UEs **104**, an Evolved Packet Core (EPC) **160**, and another core network **190** (e.g., a 5G Core (5GC)). The base stations **102** may include macrocells (high power cellular base station) and/or small cells (low power cellular base station). The macrocells include base stations. The small cells include femtocells, picocells, and microcells.

(28) The base stations **102** configured for 4G LTE (collectively referred to as Evolved Universal Mobile Telecommunications System (UMTS) Terrestrial Radio Access Network (E-UTRAN)) may interface with the EPC **160** through first backhaul links **132** (e.g., S1 interface). The base stations

102 configured for 5G NR (collectively referred to as Next Generation RAN (NG-RAN)) may interface with core network **190** through second backhaul links **184**. In addition to other functions, the base stations **102** may perform one or more of the following functions: transfer of user data, radio channel ciphering and deciphering, integrity protection, header compression, mobility control functions (e.g., handover, dual connectivity), inter-cell interference coordination, connection setup and release, load balancing, distribution for non-access stratum (NAS) messages, NAS node selection, synchronization, radio access network (RAN) sharing, multimedia broadcast multicast service (MBMS), subscriber and equipment trace, RAN information management (RIM), paging, positioning, and delivery of warning messages. The base stations **102** may communicate directly or indirectly (e.g., through the EPC **160** or core network **190**) with each other over third backhaul links **134** (e.g., X2 interface). The first backhaul links **132**, the second backhaul links **184**, and the third backhaul links **134** may be wired or wireless.

(29) The base stations **102** may wirelessly communicate with the UEs **104**. Each of the base stations **102** may provide communication coverage for a respective geographic coverage area **110**. There may be overlapping geographic coverage areas **110**. For example, the small cell **102′** may have a coverage area **110′** that overlaps the coverage area **110** of one or more macro base stations **102**. A network that includes both small cell and macrocells may be known as a heterogeneous network. A heterogeneous network may also include Home Evolved Node Bs (eNBs) (HeNBs), which may provide service to a restricted group known as a closed subscriber group (CSG). The communication links **120** between the base stations **102** and the UEs **104** may include uplink (UL) (also referred to as reverse link) transmissions from a UE **104** to a base station **102** and/or downlink (DL) (also referred to as forward link) transmissions from a base station **102** to a UE **104**. The communication links **120** may use multiple-input and multiple-output (MIMO) antenna technology, including spatial multiplexing, beamforming, and/or transmit diversity. The communication links may be through one or more carriers. The base stations **102**/UEs **104** may use spectrum up to Y MHz (e.g., 5, 10, 15, 20, 100, 400, etc. MHz) bandwidth per carrier allocated in a carrier aggregation of up to a total of Yx MHz (x component carriers) used for transmission in each direction. The carriers may or may not be adjacent to each other. Allocation of carriers may be asymmetric with respect to DL and UL (e.g., more or fewer carriers may be allocated for DL than for UL). The component carriers may include a primary component carrier and one or more secondary component carriers. A primary component carrier may be referred to as a primary cell (PCell) and a secondary component carrier may be referred to as a secondary cell (SCell).

(30) Certain UEs **104** may communicate with each other using device-to-device (D2D) communication link **158**. The D2D communication link **158** may use the DL/UL WWAN spectrum. The D2D communication link **158** may use one or more sidelink channels, such as a physical sidelink broadcast channel (PSBCH), a physical sidelink discovery channel (PSDCH), a physical sidelink shared channel (PSSCH), and a physical sidelink control channel (PSCCH). D2D communication may be through a variety of wireless D2D communications systems, such as for example, WiMedia, Bluetooth, ZigBee, Wi-Fi based on the Institute of Electrical and Electronics Engineers (IEEE) 802.11 standard, LTE, or NR.

(31) The wireless communications system may further include a Wi-Fi access point (AP) **150** in communication with Wi-Fi stations (STAs) **152** via communication links **154**, e.g., in a 5 GHz unlicensed frequency spectrum or the like. When communicating in an unlicensed frequency spectrum, the STAs **152**/AP **150** may perform a clear channel assessment (CCA) prior to communicating in order to determine whether the channel is available.

(32) The small cell **102′** may operate in a licensed and/or an unlicensed frequency spectrum. When operating in an unlicensed frequency spectrum, the small cell **102′** may employ NR and use the same unlicensed frequency spectrum (e.g., 5 GHz, or the like) as used by the Wi-Fi AP **150**. The small cell **102′**, employing NR in an unlicensed frequency spectrum, may boost coverage to and/or increase capacity of the access network.

(33) The electromagnetic spectrum is often subdivided, based on frequency/wavelength, into various classes, bands, channels, etc. In 5G NR, two initial operating bands have been identified as frequency range designations FR1 (410 MHz-7.125 GHz) and FR2 (24.25 GHz-52.6 GHz). Although a portion of FR1 is greater than 6 GHz, FR1 is often referred to (interchangeably) as a "sub-6 GHz" band in various documents and articles. A similar nomenclature issue sometimes occurs with regard to FR2, which is often referred to (interchangeably) as a "millimeter wave" band in documents and articles, despite being different from the extremely high frequency (EHF) band (30 GHz-300 GHz) which is identified by the International Telecommunications Union (ITU) as a "millimeter wave" band.
(34) The frequencies between FR1 and FR2 are often referred to as mid-band frequencies. Recent 5G NR studies have identified an operating band for these mid-band frequencies as frequency range designation FR3 (7.125 GHz-24.25 GHz). Frequency bands falling within FR3 may inherit FR1 characteristics and/or FR2 characteristics, and thus may effectively extend features of FR1 and/or FR2 into mid-band frequencies. In addition, higher frequency bands are currently being explored to extend 5G NR operation beyond 52.6 GHz. For example, three higher operating bands have been identified as frequency range designations FR2-2 (52.6 GHz-71 GHz), FR4 (52.6 GHz-114.25 GHz), and FR5 (114.25 GHz-300 GHz). Each of these higher frequency bands falls within the EHF band.
(35) With the above aspects in mind, unless specifically stated otherwise, it should be understood that the term "sub-6 GHz" or the like if used herein may broadly represent frequencies that may be less than 6 GHz, may be within FR1, or may include mid-band frequencies. Further, unless specifically stated otherwise, it should be understood that the term "millimeter wave" or the like if used herein may broadly represent frequencies that may include mid-band frequencies, may be within FR2, FR4, FR2-2, and/or FR5, or may be within the EHF band.
(36) A base station **102**, whether a small cell **102′** or a large cell (e.g., macro base station), may include and/or be referred to as an eNB, gNodeB (gNB), or another type of base station. Some base stations, such as gNB **180** may operate in a traditional sub 6 GHz spectrum, in millimeter wave frequencies, and/or near millimeter wave frequencies in communication with the UE **104**. When the gNB **180** operates in millimeter wave or near millimeter wave frequencies, the gNB **180** may be referred to as a millimeter wave base station. The millimeter wave base station **180** may utilize beamforming **182** with the UE **104** to compensate for the path loss and short range. The base station **180** and the UE **104** may each include a plurality of antennas, such as antenna elements, antenna panels, and/or antenna arrays to facilitate the beamforming.
(37) The base station **180** may transmit a beamformed signal to the UE **104** in one or more transmit directions **182′**. The UE **104** may receive the beamformed signal from the base station **180** in one or more receive directions **182″**. The UE **104** may also transmit a beamformed signal to the base station **180** in one or more transmit directions. The base station **180** may receive the beamformed signal from the UE **104** in one or more receive directions. The base station **180**/UE **104** may perform beam training to determine the best receive and transmit directions for each of the base station **180**/UE **104**. The transmit and receive directions for the base station **180** may or may not be the same. The transmit and receive directions for the UE **104** may or may not be the same.
(38) The EPC **160** may include a Mobility Management Entity (MME) **162**, other MMEs **164**, a Serving Gateway **166**, a Multimedia Broadcast Multicast Service (MBMS) Gateway **168**, a Broadcast Multicast Service Center (BM-SC) **170**, and a Packet Data Network (PDN) Gateway **172**. The MME **162** may be in communication with a Home Subscriber Server (HSS) **174**. The MME **162** is the control node that processes the signaling between the UEs **104** and the EPC **160**. Generally, the MME **162** provides bearer and connection management. All user Internet protocol (IP) packets are transferred through the Serving Gateway **166**, which itself is connected to the PDN Gateway **172**. The PDN Gateway **172** provides UE IP address allocation as well as other functions. The PDN Gateway **172** and the BM-SC **170** are connected to the IP Services **176**. The IP Services

176 may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a PS Streaming Service, and/or other IP services. The BM-SC **170** may provide functions for MBMS user service provisioning and delivery. The BM-SC **170** may serve as an entry point for content provider MBMS transmission, may be used to authorize and initiate MBMS Bearer Services within a public land mobile network (PLMN), and may be used to schedule MBMS transmissions. The MBMS Gateway **168** may be used to distribute MBMS traffic to the base stations **102** belonging to a Multicast Broadcast Single Frequency Network (MBSFN) area broadcasting a particular service, and may be responsible for session management (start/stop) and for collecting eMBMS related charging information.

(39) The core network **190** may include an Access and Mobility Management Function (AMF) **192**, other AMFs **193**, a Session Management Function (SMF) **194**, and a User Plane Function (UPF) **195**. The AMF **192** may be in communication with a Unified Data Management (UDM) **196**. The AMF **192** is the control node that processes the signaling between the UEs **104** and the core network **190**. Generally, the AMF **192** provides QoS flow and session management. All user Internet protocol (IP) packets are transferred through the UPF **195**. The UPF **195** provides UE IP address allocation as well as other functions. The UPF **195** is connected to the IP Services **197**. The IP Services **197** may include the Internet, an intranet, an IP Multimedia Subsystem (IMS), a Packet Switch (PS) Streaming (PSS) Service, and/or other IP services.

(40) The base station may include and/or be referred to as a gNB, Node B, eNB, an access point, a base transceiver station, a radio base station, a radio transceiver, a transceiver function, a basic service set (BSS), an extended service set (ESS), a transmit reception point (TRP), or some other suitable terminology. The base station **102** provides an access point to the EPC **160** or core network **190** for a UE **104**. Examples of UEs **104** include a cellular phone, a smart phone, a session initiation protocol (SIP) phone, a laptop, a personal digital assistant (PDA), a satellite radio, a global positioning system, a multimedia device, a video device, a digital audio player (e.g., MP3 player), a camera, a game console, a tablet, a smart device, a wearable device, a vehicle, an electric meter, a gas pump, a large or small kitchen appliance, a healthcare device, an implant, a sensor/actuator, a display, or any other similar functioning device. Some of the UEs **104** may be referred to as IoT devices (e.g., parking meter, gas pump, toaster, vehicles, heart monitor, etc.). The UE **104** may also be referred to as a station, a mobile station, a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable terminology. In some scenarios, the term UE may also apply to one or more companion devices such as in a device constellation arrangement. One or more of these devices may collectively access the network and/or individually access the network.

(41) Referring again to FIG. **1**, in some aspects, the UE **104** may include an access component **198**. In some aspects, the access component **198** may be configured to receive, from a base station, system information indicating a set of resources or a configuration for spectrum sensing. In some aspects, the access component **198** may be further configured to perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. In some aspects, the access component **198** may be further configured to select a RO or a cell of the one or more cells based on the acquired information about the FD operation or CLI.

(42) In certain aspects, the base station **180** may include an access component **199**. In some aspects, the access component **199** may be configured to transmit, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access. In some aspects, the access component **199** may be further configured to receive, from the UE, communication based on a selection of a

RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI.
(43) Although the following description may be focused on 5G NR, the concepts described herein may be applicable to other similar areas, such as LTE, LTE-A, CDMA, GSM, and other wireless technologies.
(44) FIG. **2**A is a diagram **200** illustrating an example of a first subframe within a 5G NR frame structure. FIG. **2**B is a diagram **230** illustrating an example of DL channels within a 5G NR subframe. FIG. **2**C is a diagram **250** illustrating an example of a second subframe within a 5G NR frame structure. FIG. **2**D is a diagram **280** illustrating an example of UL channels within a 5G NR subframe. The 5G NR frame structure may be frequency division duplexed (FDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for either DL or UL, or may be time division duplexed (TDD) in which for a particular set of subcarriers (carrier system bandwidth), subframes within the set of subcarriers are dedicated for both DL and UL. In the examples provided by FIGS. **2**A, **2**C, the 5G NR frame structure is assumed to be TDD, with subframe 4 being configured with slot format 28 (with mostly DL), where D is DL, U is UL, and F is flexible for use between DL/UL, and subframe 3 being configured with slot format 1 (with all UL). While subframes 3, 4 are shown with slot formats 1, 28, respectively, any particular subframe may be configured with any of the various available slot formats 0-61. Slot formats 0, 1 are all DL, UL, respectively. Other slot formats 2-61 include a mix of DL, UL, and flexible symbols. UEs are configured with the slot format (dynamically through DL control information (DCI), or semi-statically/statically through radio resource control (RRC) signaling) through a received slot format indicator (SFI). Note that the description infra applies also to a 5G NR frame structure that is TDD.
(45) FIGS. **2**A-**2**D illustrate a frame structure, and the aspects of the present disclosure may be applicable to other wireless communication technologies, which may have a different frame structure and/or different channels. A frame (10 ms) may be divided into 10 equally sized subframes (1 ms). Each subframe may include one or more time slots. Subframes may also include mini-slots, which may include 7, 4, or 2 symbols. Each slot may include 14 or 12 symbols, depending on whether the cyclic prefix (CP) is normal or extended. For normal CP, each slot may include 14 symbols, and for extended CP, each slot may include 12 symbols. The symbols on DL may be CP orthogonal frequency division multiplexing (OFDM) (CP-OFDM) symbols. The symbols on UL may be CP-OFDM symbols (for high throughput scenarios) or discrete Fourier transform (DFT) spread OFDM (DFT-s-OFDM) symbols (also referred to as single carrier frequency-division multiple access (SC-FDMA) symbols) (for power limited scenarios; limited to a single stream transmission). The number of slots within a subframe is based on the CP and the numerology. The numerology defines the subcarrier spacing (SCS) and, effectively, the symbol length/duration, which is equal to 1/SCS.
(46) TABLE-US-00001 SCS Cyclic μ Δf = 2.sup.μ .Math. 15 [kHz] prefix 0 15 Normal 1 30 Normal 2 60 Normal, Extended 3 120 Normal 4 240 Normal
(47) For normal CP (14 symbols/slot), different numerologies μ0 to 4 allow for 1, 2, 4, 8, and 16 slots, respectively, per subframe. For extended CP, the numerology 2 allows for 4 slots per subframe. Accordingly, for normal CP and numerology μ, there are 14 symbols/slot and 2.sup.μ slots/subframe. The subcarrier spacing may be equal to 2.sup.μ*15 kHz, where μ is the numerology 0 to 4. As such, the numerology μ=0 has a subcarrier spacing of 15 kHz and the numerology μ=4 has a subcarrier spacing of 240 kHz. The symbol length/duration is inversely related to the subcarrier spacing. FIGS. **2**A-**2**D provide an example of normal CP with 14 symbols per slot and numerology μ=2 with 4 slots per subframe. The slot duration is 0.25 ms, the subcarrier spacing is 60 kHz, and the symbol duration is approximately 16.67 μs. Within a set of frames, there may be one or more different bandwidth parts (BWPs) (see FIG. **2**B) that are frequency division multiplexed. Each BWP may have a particular numerology and CP (normal or extended).

(48) A resource grid may be used to represent the frame structure. Each time slot includes a resource block (RB) (also referred to as physical RBs (PRBs)) that extends 12 consecutive subcarriers. The resource grid is divided into multiple resource elements (REs). The number of bits carried by each RE depends on the modulation scheme.
(49) As illustrated in FIG. **2**A, some of the REs carry reference (pilot) signals (RS) for the UE. The RS may include demodulation RS (DM-RS) (indicated as R for one particular configuration, but other DM-RS configurations are possible) and channel state information reference signals (CSI-RS) for channel estimation at the UE. The RS may also include beam measurement RS (BRS), beam refinement RS (BRRS), and phase tracking RS (PT-RS).
(50) FIG. **2**B illustrates an example of various DL channels within a subframe of a frame. The physical downlink control channel (PDCCH) carries DCI within one or more control channel elements (CCEs) (e.g., 1, 2, 4, 8, or 16 CCEs), each CCE including six RE groups (REGs), each REG including 12 consecutive REs in an OFDM symbol of an RB. A PDCCH within one BWP may be referred to as a control resource set (CORESET). A UE is configured to monitor PDCCH candidates in a PDCCH search space (e.g., common search space, UE-specific search space) during PDCCH monitoring occasions on the CORESET, where the PDCCH candidates have different DCI formats and different aggregation levels. Additional BWPs may be located at greater and/or lower frequencies across the channel bandwidth. A primary synchronization signal (PSS) may be within symbol 2 of particular subframes of a frame. The PSS is used by a UE **104** to determine subframe/symbol timing and a physical layer identity. A secondary synchronization signal (SSS) may be within symbol 4 of particular subframes of a frame. The SSS is used by a UE to determine a physical layer cell identity group number and radio frame timing. Based on the physical layer identity and the physical layer cell identity group number, the UE can determine a physical cell identifier (PCI). Based on the PCI, the UE can determine the locations of the DM-RS. The physical broadcast channel (PBCH), which carries a master information block (MIB), may be logically grouped with the PSS and SSS to form a synchronization signal (SS)/PBCH block (also referred to as SS block (SSB)). The MIB provides a number of RBs in the system bandwidth and a system frame number (SFN). The physical downlink shared channel (PDSCH) carries user data, broadcast system information not transmitted through the PBCH such as system information blocks (SIBs), and paging messages.
(51) As illustrated in FIG. **2**C, some of the REs carry DM-RS (indicated as R for one particular configuration, but other DM-RS configurations are possible) for channel estimation at the base station. The UE may transmit DM-RS for the physical uplink control channel (PUCCH) and DM-RS for the physical uplink shared channel (PUSCH). The PUSCH DM-RS may be transmitted in the first one or two symbols of the PUSCH. The PUCCH DM-RS may be transmitted in different configurations depending on whether short or long PUCCHs are transmitted and depending on the particular PUCCH format used. The UE may transmit sounding reference signals (SRS). The SRS may be transmitted in the last symbol of a subframe. The SRS may have a comb structure, and a UE may transmit SRS on one of the combs. The SRS may be used by a base station for channel quality estimation to enable frequency-dependent scheduling on the UL.
(52) FIG. **2**D illustrates an example of various UL channels within a subframe of a frame. The PUCCH may be located as indicated in one configuration. The PUCCH carries uplink control information (UCI), such as scheduling requests, a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indicator (RI), and hybrid automatic repeat request (HARQ) acknowledgment (ACK) (HARQ-ACK) feedback (i.e., one or more HARQ ACK bits indicating one or more ACK and/or negative ACK (NACK)). The PUSCH carries data, and may additionally be used to carry a buffer status report (BSR), a power headroom report (PHR), and/or UCI.
(53) FIG. **3** is a block diagram of a base station **310** in communication with a UE **350** in an access network. In the DL, IP packets from the EPC **160** may be provided to a controller/processor **375**. The controller/processor **375** implements layer 3 and layer 2 functionality. Layer 3 includes a radio

resource control (RRC) layer, and layer 2 includes a service data adaptation protocol (SDAP) layer, a packet data convergence protocol (PDCP) layer, a radio link control (RLC) layer, and a medium access control (MAC) layer. The controller/processor **375** provides RRC layer functionality associated with broadcasting of system information (e.g., MIB, SIBs), RRC connection control (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), inter radio access technology (RAT) mobility, and measurement configuration for UE measurement reporting; PDCP layer functionality associated with header compression/decompression, security (ciphering, deciphering, integrity protection, integrity verification), and handover support functions; RLC layer functionality associated with the transfer of upper layer packet data units (PDUs), error correction through ARQ, concatenation, segmentation, and reassembly of RLC service data units (SDUs), re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto transport blocks (TBs), demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.
(54) The transmit (TX) processor **316** and the receive (RX) processor **370** implement layer 1 functionality associated with various signal processing functions. Layer 1, which includes a physical (PHY) layer, may include error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, interleaving, rate matching, mapping onto physical channels, modulation/demodulation of physical channels, and MIMO antenna processing. The TX processor **316** handles mapping to signal constellations based on various modulation schemes (e.g., binary phase-shift keying (BPSK), quadrature phase-shift keying (QPSK), M-phase-shift keying (M-PSK), M-quadrature amplitude modulation (M-QAM)). The coded and modulated symbols may then be split into parallel streams. Each stream may then be mapped to an OFDM subcarrier, multiplexed with a reference signal (e.g., pilot) in the time and/or frequency domain, and then combined together using an Inverse Fast Fourier Transform (IFFT) to produce a physical channel carrying a time domain OFDM symbol stream. The OFDM stream is spatially precoded to produce multiple spatial streams. Channel estimates from a channel estimator **374** may be used to determine the coding and modulation scheme, as well as for spatial processing. The channel estimate may be derived from a reference signal and/or channel condition feedback transmitted by the UE **350**. Each spatial stream may then be provided to a different antenna **320** via a separate transmitter **318** TX. Each transmitter **318** TX may modulate a radio frequency (RF) carrier with a respective spatial stream for transmission.
(55) At the UE **350**, each receiver **354** RX receives a signal through its respective antenna **352**. Each receiver **354** RX recovers information modulated onto an RF carrier and provides the information to the receive (RX) processor **356**. The TX processor **368** and the RX processor **356** implement layer 1 functionality associated with various signal processing functions. The RX processor **356** may perform spatial processing on the information to recover any spatial streams destined for the UE **350**. If multiple spatial streams are destined for the UE **350**, they may be combined by the RX processor **356** into a single OFDM symbol stream. The RX processor **356** then converts the OFDM symbol stream from the time-domain to the frequency domain using a Fast Fourier Transform (FFT). The frequency domain signal comprises a separate OFDM symbol stream for each subcarrier of the OFDM signal. The symbols on each subcarrier, and the reference signal, are recovered and demodulated by determining the most likely signal constellation points transmitted by the base station **310**. These soft decisions may be based on channel estimates computed by the channel estimator **358**. The soft decisions are then decoded and deinterleaved to recover the data and control signals that were originally transmitted by the base station **310** on the physical channel. The data and control signals are then provided to the controller/processor **359**, which implements layer 3 and layer 2 functionality.
(56) The controller/processor **359** can be associated with a memory **360** that stores program codes

and data. The memory **360** may be referred to as a computer-readable medium. In the UL, the controller/processor **359** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, and control signal processing to recover IP packets from the EPC **160**. The controller/processor **359** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(57) Similar to the functionality described in connection with the DL transmission by the base station **310**, the controller/processor **359** provides RRC layer functionality associated with system information (e.g., MIB, SIBs) acquisition, RRC connections, and measurement reporting; PDCP layer functionality associated with header compression/decompression, and security (ciphering, deciphering, integrity protection, integrity verification); RLC layer functionality associated with the transfer of upper layer PDUs, error correction through ARQ, concatenation, segmentation, and reassembly of RLC SDUs, re-segmentation of RLC data PDUs, and reordering of RLC data PDUs; and MAC layer functionality associated with mapping between logical channels and transport channels, multiplexing of MAC SDUs onto TBs, demultiplexing of MAC SDUs from TBs, scheduling information reporting, error correction through HARQ, priority handling, and logical channel prioritization.

(58) Channel estimates derived by a channel estimator **358** from a reference signal or feedback transmitted by the base station **310** may be used by the TX processor **368** to select the appropriate coding and modulation schemes, and to facilitate spatial processing. The spatial streams generated by the TX processor **368** may be provided to different antenna **352** via separate transmitters **354**TX. Each transmitter **354**TX may modulate an RF carrier with a respective spatial stream for transmission.

(59) The UL transmission is processed at the base station **310** in a manner similar to that described in connection with the receiver function at the UE **350**. Each receiver **318**RX receives a signal through its respective antenna **320**. Each receiver **318**RX recovers information modulated onto an RF carrier and provides the information to a RX processor **370**.

(60) The controller/processor **375** can be associated with a memory **376** that stores program codes and data. The memory **376** may be referred to as a computer-readable medium. In the UL, the controller/processor **375** provides demultiplexing between transport and logical channels, packet reassembly, deciphering, header decompression, control signal processing to recover IP packets from the UE **350**. IP packets from the controller/processor **375** may be provided to the EPC **160**. The controller/processor **375** is also responsible for error detection using an ACK and/or NACK protocol to support HARQ operations.

(61) At least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359** may be configured to perform aspects in connection with access component **198** of FIG. **1**.

(62) At least one of the TX processor **316**, the RX processor **370**, and the controller/processor **375** may be configured to perform aspects in connection with access component **199** of FIG. **1**.

(63) In some wireless communication systems, FD capability (supporting simultaneous UL or DL transmission) may be present at the base station, the UE, or both the base station and the UE. For example, at the UE, UL transmissions may be transmitted from a first panel of the UE while simultaneous DL receptions may be received at a second panel of the UE. The first panel and the second panel may be different panels of the antenna(s) on the UE. As another example, at the base station, UL receptions may be received from a first panel of the base station while simultaneous DL transmissions may be transmitted at a second panel of the base station. The first panel and the second panel may be different panels of the antenna(s) on the base station.

(64) FIG. **4**A illustrates an example **400** of in-band full-duplex (IBFD) resources (which may also be referred to as single-frequency full-duplex). FIG. **4**B illustrates an example **410** of IBFD resources. FIG. **4**C illustrates an example **420** of sub-band full-duplex resources. In IBFD, signals may be transmitted and received in overlapping times and overlapping in frequency. As shown in the example **400** illustrated in FIG. **4**A, a time and a frequency allocation of transmission resources

402 may fully overlap with a time and a frequency allocation of reception resources **404**. In the example **410** illustrated in FIG. **4**B, a time and a frequency allocation of transmission resources **412** may partially overlap with a time and a frequency of allocation of reception resources **414**.
(65) IBFD is in contrast to sub-band frequency division duplex (FDD), where transmission and reception resources may overlap in time using different frequencies, as shown in the example **420** illustrated in FIG. **4**A. In the example **420**, the transmission resources **422** are separated from the reception resources **424** by a guard band **426**. The guard band may be frequency resources, or a gap in frequency resources, provided between the transmission resources **422** and the reception resources **424**. Separating the frequency resources for transmission and reception with a guard band may help to reduce self-interference. Transmission and reception resources that are immediately adjacent to each other may be considered as having a guard band width of 0. As an output signal, e.g., from a UE transmitter may extends outside the transmission resources, the guard band may reduce interference experienced by the UE. Sub-band FDD may also be referred to as "flexible duplex."
(66) As previously described, an FD network node, such as a base station or UE in the cellular network, may communicate simultaneously in UL and DL with two half-duplex panels using the same radio resources. Due to the simultaneous Tx/Rx nature of FD communication, a UE may experience self-interference caused by signal leakage from the transmitting panel to the receiving panel. Such interference (e.g., self-interference or interference caused by other devices) may impact the quality of the communication or even lead to a loss of information. Therefore, beam separations and other mitigation methods may be used to support FD capability. Cluster echo (which may be otherwise referred to as "cluster interference") from surrounding objects may also be present. Therefore, a UE capable of FD communication may not always work in FD mode due to the potentially high interference.
(67) By supporting FD, latency of communications may be potentially reduced. For example, it may be possible for a UE to receive DL signal in slots assigned for UL, which may enable latency savings. Furthermore, by supporting FD, spectrum efficiency per cell and per UE may be improved because resource utilization over the spectrum may be more efficient.
(68) FIGS. **5**A-C illustrate various use cases of FD communication. Referring to FIG. **5**A, as shown in example **500**, an FD UE **502** may be simultaneously connected to a first TRP **504**A and a second TRP **504**B. The FD UE **502** may be receiving DL communications from the first TRP **504**A while transmitting UL communications to the second TRP **504**B. In some aspects, the first TRP **504**A and the second TRP **504**B may be half-duplex (HD) and may not support FD operations. In some aspects, the first TRP **504**A and the second TRP **504**B may support FD operations.
(69) Referring to FIG. **5**B, as shown in example **510**, an FD TRP **514** of a base station may be transmitting DL communications to a first UE **512**A while receiving UL communications from a second UE **512**B. In some aspects, the first UE **512**A and the second UE **512**B may be HD and may not support FD operations. In some aspects, the first UE **512**A and the second UE **512**B may support FD operations.
(70) Referring to FIG. **5**C, as shown in example **520**, an FD TRP **524** of a base station may be transmitting DL communications to an FD UE **522** while receiving UL communications from the FD UE **522**. The FD UE **522** may be transmitting UL communications to the FD TRP **524** while receiving DL communications from the FD TRP **524**.
(71) For an FD TRP of a base station communication with a first UE and a second UE that may be both HD (e.g., example **510**), the HD UEs may benefit from the potential advantages of latency saving, better spectrum efficiency, or other efficiencies resulting from the base station operating in FD. On the other hand, the HD UEs may also experience extra CLI between the HD UEs due to the FD operations of the TRP. For example, during a slot where the first UE **512**A receives DL communications, the nearby second UE **512**B may be sending UL communications to the TRP, which may potentially cause CLI to the first UE **512**A.

(72) There are different ways to address the potential CLI. For example, the network may configure CLI resources for the UEs to measure and report CLI. After receiving reports on CLI, by pairing the first UE and the second UE and accordingly scheduling the UL/DL communications for the first UE and the second UE, the base station may operation in FD and simultaneously serve DL and UL UEs with minor CLI. However, sometimes CLI may be present in initial access (e.g., when ROs for one UE overlaps with DL communications for another UE). Therefore, the network may configure cell-specific CLI resources (e.g., in a broadcast manner), during which connected UEs may be configured to send SRS. An initial access UE may measure CLI from neighboring UE and decide whether the initial access UE may use a full-duplex RO (overlapping with DL reception of nearby UEs) without causing excessive CLI to other nearby UEs or not based on the measurement. The SRS may be periodic and so that the initial access UEs may be able to measure CLI and accordingly select FD RO or HD FO. However, such SRS may introduce overhead for the connected UEs sending the SRS. Aspects provided herein may use spectrum sensing during the initial access for cell selection or RO selection, preventing overhead for the connected UEs while resolving the potential CLI issue for initial access UEs. Spectrum sensing may be the process of observing the frequency bands and sensing activities originated from other devices in the frequency band. In some aspects, the initial access UE may perform spectrum sensing during the initial access to acquire information about the FD operation or CLI profile of a cell.

(73) In some aspects of wireless communications, such as during initial access, a UE may use a RACH procedure in order to communicate with a base station. For example, the UE may use the random access procedure to request a radio resource control (RRC) connection, to re-establish an RRC connection, to resume an RRC connection, etc. RACH procedures may include a number of different types of random access procedures, e.g., contention based random access (CBRA) may be performed when a UE is not synchronized with a base station and contention free random access (CFRA) may be performed when a UE was previously synchronized with a base station. In CBRA, a UE may randomly select a RACH preamble sequence, e.g., from a set of RACH preamble sequences. As the UE randomly selects the RACH preamble sequence, the base station may receive another RACH preamble from a different UE at the same time. Thus, CBRA provides for the base station with the ability to resolve such contention among multiple UEs. In CFRA, the network may allocate a RACH preamble sequence to the UE rather than the UE randomly selecting a RACH preamble sequence. This may help to avoid potential collisions with a RACH preamble from another UE using the same sequence. Thus, CFRA may be referred to as “contention free” random access.

(74) FIG. **6** is a diagram **600** illustrating an example RACH procedure between a UE **602** and a base station **604**. In some aspects, the UE **602** may be an initial access UE and may obtain random access parameters (which may be otherwise referred to as PRACH configurations), e.g., including RACH preamble format parameters, PRACH resources (in the form of time and frequency), parameters for determining root sequences and/or cyclic shifts for a RACH preamble, etc., in system information **606** from the base station **604**. In some aspects, the system information **606** may further indicate one or more resources and a configuration for the UE **602** to perform spectrum sensing at **608**.

(75) In some aspects, the system information **606** may indicate time-domain resources, such as a set of UL or FD symbols or slots, that may be used for spectrum sensing. In some aspects, the system information **606** may indicate symbols or slots that may be used for spectrum sensing implicitly. For example, the system information **606** may indicate that all UL symbols, all flexible (that could be used for UL and DL) symbols, or all FD symbols indicated in a time division duplex (TDD) configuration may be used for spectrum sensing.

(76) In some aspects, the system information **606** may indicate frequency domain resources that may be used for performing spectrum sensing at **608**. For example, the frequency domain resources may be an initial UL bandwidth part (BWP), an indicated BWP for FD operations, or another

defined BWP. In some aspects, the frequency domain resources may be a collection of staggered resource block (RB) groups. In some aspects, the frequency domain resources may be different for different time-domain symbols/slots. For example, for a first slot/symbol, an initial UL BWP may be indicated to be used for spectrum sensing. For a second slot/symbol, an indicated BWP for FD operations may be indicated to be used for spectrum sensing.

(77) In some aspects, the system information **606** may indicate a window prior to one or more ROs that may carry RACH preambles within which the indicated UL/FD resources may be sensed. In some aspects, the system information **606** may configure one or more conditions for selecting a RO. The one or more conditions may include a threshold on measured (e.g., averaged) received signal strength indicator (RSSI) or a highest measured RSSI, or a number of resources where measured RSSI being above a threshold, or other statistics related to measurements that may be performed during spectrum sensing at **608**.

(78) In some aspects, the one or more resources that may be indicated by the system information **606** to be for spectrum sensing for the UE **602** may be used by connected UEs, such as a UE **650**, to exchange communications **652** such as transmitting UL transmissions with the base station **604**. In some aspects, there may be no extra UL signals for CLI measurements for the spectrum sensing at **608**. In some aspects, the connected UEs, such as the UE **650**, may transmit UL communications normally without considering that the resources may be used by the UE **602** for spectrum sensing. In some aspects, based on the measured RSSI during the spectrum sensing at **608**, the UE **602** may perform RO selection **609** based on the one or more conditions. In some aspects, the UE **602** may perform RO selection **609** based on a set of conditions configured for/determined by the UE without base station signaling.

(79) In some aspects, based on the spectrum sensing at **608**, the UE **602** may determine the profile and severity of CLI. Based on the profile and severity of CLI, the UE **602** may also perform cell selection at **609**. For example, the UE may have detected two cells and may acquire more information during the spectrum sensing at **608** to sub-select one of the two cells. In some aspects, an FD cell or another cell with less chance of high CLI may be selected by the UE **602** for further transmission of random access messages. In some aspects, the UE **602** may select the cell based on criteria configured for/determined by the UE without base station signaling.

(80) In some aspects, the system information **606** may not explicitly indicate information about resources that may be used for CLI measurement. In such aspects, the UE **602** may, based on TDD and BWP configuration in the system information **606**, select resources for the spectrum sensing at **608**. In some aspects, the UE **602** may acquire information about FD resources (e.g., based on TDD indication in remaining minimum system information (RMSI)). Then the UE **602** may decide to perform spectrum sensing within the FD resources at **608**.

(81) In some aspects, the spectrum sensing at **608** may include RSSI measurement and further processing based on RSSI measurements. For example, measured RSSI within resources with both DL and UL communications may include total reception power from both the base station **604** and nearby UEs. Therefore, the UE **602** may not differentiate and separately measure CLI caused by the UEs. The spectrum sensing at **608** may include further processing to estimate the CLI caused by the UEs. For example, in some aspects, the UE may separately measure (e.g., and may average) RSSI over DL resources that do not have UL transmissions. Based on the RSSI over the DL resources that do not have UL transmissions and a total measured RSSI over full-duplex resources, the UE **602** may estimate average CLI-RSSI caused by other UEs' UL transmissions. For example, total RSSI minus DL RSSI may be equal to an estimate of UL/CLI RSSI caused by other UEs' UL transmissions.

(82) As another example, in some aspects, the UE **602** may assume that DL transmission may spans wider bandwidth (BW) or may be associated with a more uniform power spectral density (PSD). The UE **602** may identify and distinguish UL from DL based on the measured spectrum. Further details of example measured spectrums are described in connection with FIGS. **7**A-**7**C. As another

example, in some aspects, the UE **602** may not have prior information about FD resources. For example, based on spectrum sensing and evaluating the estimated power spectrum, the UE **602** may determine the time resources where both DL and UL present (i.e., measured spectrum matches a mixture of DL and UL), and may estimate the UL/CLI RSSI caused by other UEs' UL transmissions accordingly. In another example, in some aspects, the UE **602** may estimate whether a cell is an FD cell based on whether both DL and UL are present. The UE **602** may accordingly select a cell estimated to be FD that may support FD operation or select another cell.

(83) After selecting RO(s) or cells, the UE **602** may initiate the random access message exchange by sending, to the base station **604**, a first random access message **610** (e.g., message 1 (Msg 1)) including a RACH preamble. In some aspects, multiple short preambles may be multiplexed in time within a single RO. Prior to sending the first random access message **610**, the UE may configure a RACH preamble parameter including a cyclic prefix (CP) and a preamble sequence. Each of the preamble formats may be associated with a different CP and a different preamble sequence. A preamble format may be grouped into two categories: long preamble and short preamble. By way of example, a long preamble may last for more than 1 ms in the time domain and a short preamble may last for less than 1 ms in the time domain. For example, a long preamble may be based on a sequence length of L=839. A long preamble may include preamble format 0, format 1, format 2, and format 3. By way of example, an SCS associated with a long preamble may be 1.25 kHz or 5 kHz. A long preamble may be used for the FR1 frequency band. A long preamble with 1.25 kHz SCS may occupy, by way of example, 6 resource blocks in the frequency domain. A long preamble with 5 kHz may occupy, by way of example, 24 resource blocks in the frequency domain.

(84) A RACH preamble may be transmitted with an identifier, such as a random access RNTI (RA-RNTI). The UE **602** may randomly select a RACH preamble sequence, e.g., from a set of RACH preamble sequences corresponding to the preamble formats. If the UE **602** randomly selects the RACH preamble sequence, the base station **604** may receive another RACH preamble from a different UE at the same time. In some examples, a RACH preamble sequence may be assigned to the UE **602**.

(85) The base station may respond to the first random access message **610** by sending a second random access message **612** (e.g., Msg 2 or Msg2) using a PDSCH and including a random access response (RAR). The RAR may include, e.g., an identifier of the RACH preamble sent by the UE, a timing advance (TA), an uplink grant for the UE to transmit data, a cell radio network temporary identifier (C-RNTI) or other identifier, and/or a back-off indicator. Upon receiving the RAR (e.g., **612**), the UE **602** may transmit a third random access message **614** (e.g., Msg 3 or Msg3) to the base station **604**, e.g., using a PUSCH, that may include an RRC connection request, an RRC connection re-establishment request, or an RRC connection resume request, depending on the trigger for initiating the random access procedure. The base station **604** may then complete the random access procedure by sending a fourth random access message **616** (e.g., Msg 4 or Msg4) to the UE **602**, e.g., using a PDCCH for scheduling and a PDSCH for the message. The fourth random access message **616** may include a random access response message that includes timing advancement information, contention resolution information, and/or RRC connection setup information. The UE **602** may monitor for a PDCCH, e.g., with the C-RNTI. If the PDCCH is successfully decoded, the UE **602** may also decode a PDSCH. The UE **602** may send HARQ feedback for any data carried in the fourth random access message.

(86) In some wireless communication systems, the UE **602** may request a Msg 3 PUSCH repetition. For example, the UE **602** may request a Msg 3 PUSCH repetition to improve coverage. In some wireless communication systems, the UE **602** may request the Msg 3 PUSCH repetition via PRACH resources. In some aspects, the UE **602** may indicate the request via RACH occasions (ROs). In some aspects, the UE **602** may not indicate the request via ROs. The UE **602** may request the Msg 3 PUSCH repetition based on different criteria, such as a synchronization signal (SS)

reference signal received power (RSRP). The UE **602** may also transmit one or more PRACH repetitions using the PRACH resources that may carry requests for Msg 3 PUSCH repetitions. Aspects provided herein may enable implicitly indicating Msg 3 PUSCH repetitions via PRACH repetitions, which may improve the efficiency of a RACH procedure between a UE and a base station.

(87) FIGS. **7**A-C illustrate example sensed spectrums. As illustrated in example **700** of FIG. **7**A, a DL spectrum measured based on resources that may be used for DL without being used for UL may be flat and wide. As illustrated in example **730** of FIG. **7**B, a UL spectrum measured based on resources that may be used for UL without being used for DL may be spiky. As illustrated in example **750** of FIG. **7**C, a mixture UL and DL spectrum measured based on resources that may be used for UL and DL may exhibit characteristics of both DL and UL spectrums.

(88) FIG. **8** is a flowchart **800** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, the UE **602**; the apparatus **1102**).

(89) At **804**, the UE may perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. For example, the UE **602** may perform the spectrum sensing at **608** based on the set of resources or the configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. In some aspects, **804** may be performed by access component **1142** in FIG. **11**.

(90) At **806**, the UE may select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI. For example, the UE **602** may select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI at **609**. In some aspects, **806** may be performed by access component **1142** in FIG. **11**.

(91) FIG. **9** is a flowchart **900** of a method of wireless communication. The method may be performed by a UE (e.g., the UE **104**, the UE **602**; the apparatus **1102**).

(92) At **902**, the UE may receive, from a base station, system information indicating a set of resources or a configuration for spectrum sensing. For example, the UE **602** may receive, from a base station **604**, system information **606** indicating a set of resources or a configuration for spectrum sensing at **608**. In some aspects, **902** may be performed by access component **1142** in FIG. **11**. In some aspects, the set of resources may include one or more time domain resources including one or more UL or FD symbols or slots associated with the spectrum sensing. In some aspects, the one or more UL or FD symbols or slots associated with the spectrum sensing may include all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a TDD configuration. In some aspects, the set of resources may include one or more frequency domain resources. In some aspects, the one or more frequency domain resources may include an initial UL BWP. In some aspects, the one or more frequency domain resources may include a dedicated BWP for the FD operation. In some aspects, the one or more frequency domain resources may include a collection of staggered RB groups. In some aspects, the one or more frequency domain resources may include at least one different frequency domain resource associated with at least one different time domain resource. In some aspects, the set of resources may include a time window prior to the RO. In some aspects, the configuration may include one or more conditions associated with a selection of the RO or the one or more cells (e.g., one cell of the one or more cells). In some aspects, the one or more conditions may include one or more of: an average RSSI threshold, a highest RSSI threshold, or a number of resources associated with an RSSI above a threshold. In some aspects, the set of resources may overlap with a set of resources configured for UL communication for a connected UE. In some aspects, the one or more FD resources may include one or more TDD indications in RMSI.

(93) At **904**, the UE may perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells

of the base station during initial access. For example, the UE **602** may perform the spectrum sensing at **608** based on the set of resources or the configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access. In some aspects, **904** may be performed by access component **1142** in FIG. **11**. In some aspects, as part of **904**, the UE may, at **904**A, measure at least one total RSSI associated with at least one resource of the set of resources. In some aspects, as part of **904**, the UE may, at **904**B, measure at least one DL RSSI associated with one or more DL resources of the set of resources. In some aspects, as part of **904**, the UE may, at **904**C, estimate at least one CLI RSSI based on the at least one DL RSSI and the total RSSI. In some aspects, at **905**, which may be part of **904**, the UE may identify and distinguish an UL spectrum from a DL spectrum associated with the at least one resource. In some aspects, the DL spectrum may span a wider BW and may be associated with a more uniform PSD compared to the UL spectrum.
(94) At **906**, the UE may select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI. For example, the UE **602** may select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI at **609**. In some aspects, **906** may be performed by access component **1142** in FIG. **11**. In some aspects, the cell of the one or more cells may be a cell with a reduced chance of a high CLI. In some aspects, the set of resources may be selected based on the configuration, the configuration being a TDD configuration or a BWP configuration. In some aspects, the information about the FD operation or CLI may be based on one or more FD resources. In some aspects, as part of **906**, at **908**, the UE may determine that the at least one resource may include at least one UL resource and at least one DL resource. In some aspects, as part of **906**, at **910**, the UE may determine that a cell associated with the at least one resource is an FD cell.
(95) FIG. **10** is a flowchart **1000** of a method of wireless communication. The method may be performed by a base station (e.g., the base station **102**/**180**, the base station **604**; the apparatus **1202**).
(96) At **1002**, the base station may transmit, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access. For example, the base station **604** may transmit, to a UE **602**, system information **606** indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. In some aspects, **1002** may be performed by access component **1242** in FIG. **12**. In some aspects, the set of resources may include one or more time domain resources including one or more UL or FD symbols or slots associated with the spectrum sensing. In some aspects, the one or more UL or FD symbols or slots associated with the spectrum sensing may include all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a TDD configuration. In some aspects, the set of resources may include one or more frequency domain resources. In some aspects, the one or more frequency domain resources may include an initial UL BWP. In some aspects, the one or more frequency domain resources may include a dedicated BWP for the FD operation. In some aspects, the one or more frequency domain resources may include a collection of staggered RB groups. In some aspects, the one or more frequency domain resources may include at least one different frequency domain resource associated with at least one different time domain resource. In some aspects, the set of resources may include a time window prior to the RO. In some aspects, the configuration may include one or more conditions associated with a selection of the RO or the one or more cells (e.g., one cell of the one or more cells). In some aspects, the one or more conditions may include one or more of: an average RSSI threshold, a highest RSSI threshold, or a number of resources associated with an RSSI above a threshold. In some aspects, the set of resources may overlap with a set of resources configured for UL communication for a connected UE. In some

aspects, the one or more FD resources may include one or more TDD indications in RMSI.
(97) At **1004**, the base station may receive, from the UE, communication based on a selection of a RO or a cell of the one or more cells based on the information about at least one of FD operation or CLI. For example, the base station **604** may receive, from the UE **602**, communication (e.g., the Msg 1 **610**) based on a selection of a RO or a cell of the one or more cells based on the FD operation or CLI (e.g., at **609**). In some aspects, **1004** may be performed by access component **1242** in FIG. **12**.
(98) FIG. **11** is a diagram **1100** illustrating an example of a hardware implementation for an apparatus **1102**. The apparatus **1102** may be a UE, a component of a UE, or may implement UE functionality. In some aspects, the apparatus **1102** may include a cellular baseband processor **1104** (also referred to as a modem) coupled to a cellular RF transceiver **1122**. In some aspects, the apparatus **1102** may further include one or more subscriber identity modules (SIM) cards **1120**, an application processor **1106** coupled to a secure digital (SD) card **1108** and a screen **1110**, a Bluetooth module **1112**, a wireless local area network (WLAN) module **1114**, a Global Positioning System (GPS) module **1116**, or a power supply **1118**. The cellular baseband processor **1104** communicates through the cellular RF transceiver **1122** with the UE **104** and/or BS **102**/**180**. The cellular baseband processor **1104** may include a computer-readable medium/memory. The computer-readable medium/memory may be non-transitory. The cellular baseband processor **1104** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the cellular baseband processor **1104**, causes the cellular baseband processor **1104** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the cellular baseband processor **1104** when executing software. The cellular baseband processor **1104** further includes a reception component **1130**, a communication manager **1132**, and a transmission component **1134**. The communication manager **1132** includes the one or more illustrated components. The components within the communication manager **1132** may be stored in the computer-readable medium/memory and/or configured as hardware within the cellular baseband processor **1104**. The cellular baseband processor **1104** may be a component of the UE **350** and may include the memory **360** and/or at least one of the TX processor **368**, the RX processor **356**, and the controller/processor **359**. In one configuration, the apparatus **1102** may be a modem chip and include just the baseband processor **1104**, and in another configuration, the apparatus **1102** may be the entire UE (e.g., see **350** of FIG. **3**) and include the additional modules of the apparatus **1102**.
(99) The communication manager **1132** may include an access component **1142** that is configured to receive, from a base station, system information indicating a set of resources or a configuration for spectrum sensing, e.g., as described in connection with **802** in FIGS. **8** and **902** in FIG. **9**. The access component **1142** may be further configured to perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access, e.g., as described in connection with **804** in FIGS. **8** and **904** in FIG. **9**. The access component **1142** may be further configured to select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI, e.g., as described in connection with **806** in FIGS. **8** and **906** in FIG. **9**. The access component **1142** may be further configured to identify and distinguish an UL spectrum from a DL spectrum associated with the at least one resource, determine that the at least one resource includes at least one UL resource and at least one DL resource, or determine that a cell associated with the at least one resource is an FD cell, e.g., as described in connection with **905**, **908**, and **910** in FIG. **9**.
(100) The apparatus may include additional components that perform each of the blocks of the algorithm in the flowcharts of FIGS. **8-9**. As such, each block in the flowcharts of FIGS. **8-9** may be performed by a component and the apparatus may include one or more of those components. The components may be one or more hardware components specifically configured to carry out the

stated processes/algorithm, implemented by a processor configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by a processor, or some combination thereof.

(101) As shown, the apparatus **1102** may include a variety of components configured for various functions. In one configuration, the apparatus **1102**, and in particular the cellular baseband processor **1104**, may include means for receiving, from a base station, system information indicating a set of resources or a configuration for spectrum sensing. The cellular baseband processor **1104** may further include means for performing a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access. The cellular baseband processor **1104** may further include means for selecting a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI. The cellular baseband processor **1104** may further include means for measuring at least one total RSSI associated with at least one resource of the set of resources. The cellular baseband processor **1104** may further include means for measuring at least one DL RSSI associated with one or more DL resources of the set of resources. The cellular baseband processor **1104** may further include means for estimating at least one CLI RSSI based on the at least one DL RSSI and the total RSSI. The cellular baseband processor **1104** may further include means for identifying and distinguishing an UL spectrum from a DL spectrum associated with the at least one resource. The cellular baseband processor **1104** may further include means for determining that the at least one resource includes at least one UL resource and at least one DL resource. The cellular baseband processor **1104** may further include means for determining that a cell associated with the at least one resource is an FD cell. The means may be one or more of the components of the apparatus **1102** configured to perform the functions recited by the means. As described supra, the apparatus **1102** may include the TX Processor **368**, the RX Processor **356**, and the controller/processor **359**. As such, in one configuration, the means may be the TX Processor **368**, the RX Processor **356**, and the controller/processor **359** configured to perform the functions recited by the means.

(102) FIG. **12** is a diagram **1200** illustrating an example of a hardware implementation for an apparatus **1202**. The apparatus **1202** may be a base station, a component of a base station, or may implement base station functionality. In some aspects, the apparatus **1102** may include a baseband unit **1204**. The baseband unit **1204** may communicate through a cellular RF transceiver **1222** with the UE **104**. The baseband unit **1204** may include a computer-readable medium/memory. The baseband unit **1204** is responsible for general processing, including the execution of software stored on the computer-readable medium/memory. The software, when executed by the baseband unit **1204**, causes the baseband unit **1204** to perform the various functions described supra. The computer-readable medium/memory may also be used for storing data that is manipulated by the baseband unit **1204** when executing software. The baseband unit **1204** further includes a reception component **1230**, a communication manager **1232**, and a transmission component **1234**. The communication manager **1232** includes the one or more illustrated components. The components within the communication manager **1232** may be stored in the computer-readable medium/memory and/or configured as hardware within the baseband unit **1204**. The baseband unit **1204** may be a component of the base station **310** and may include the memory **376** and/or at least one of the TX processor **316**, the RX processor **370**, and the controller/processor **375**.

(103) The communication manager **1232** may include an access component **1242** that may transmit, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access or receive, from the UE, communication based on a selection of a RO or a cell of the one or more cells based on the information about at least one of FD operation or CLI, e.g., as described in connection with **1002** and **1004** in FIG. **10**.

(104) The apparatus may include additional components that perform each of the blocks of the algorithm in the flowchart of FIG. **10**. As such, each block in the flowcharts of FIG. **10** may be performed by a component and the apparatus may include one or more of those components. The components may be one or more hardware components specifically configured to carry out the stated processes/algorithm, implemented by a processor configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by a processor, or some combination thereof.
(105) As shown, the apparatus **1202** may include a variety of components configured for various functions. In one configuration, the apparatus **1202**, and in particular the baseband unit **1204**, may include means for transmitting, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access. The baseband unit **1204** may further include means for receiving, from the UE, communication based on a selection of a RO or a cell of the one or more cells based on the information about at least one of FD operation or CLI. The means may be one or more of the components of the apparatus **1202** configured to perform the functions recited by the means. As described supra, the apparatus **1202** may include the TX Processor **316**, the RX Processor **370**, and the controller/processor **375**. As such, in one configuration, the means may be the TX Processor **316**, the RX Processor **370**, and the controller/processor **375** configured to perform the functions recited by the means.
(106) For an FD TRP of a base station communication with a first UE and a second UE that may be both HD, the HD UEs may benefit from the potential advantages of latency saving, better spectrum efficiency, or other efficiencies resulting from the base station operating in FD. On the other hand, the HD UEs may also experience extra CLI between the HD UEs due to the FD operations of the TRP. For example, during a slot where the first UE receives DL communications, the nearby second UE may be sending UL communications to the TRP, which may potentially cause CLI to the first UE. Aspects provided herein may use spectrum sensing during the initial access for cell selection or RO selection.
(107) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of example approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not meant to be limited to the specific order or hierarchy presented.
(108) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” Terms such as “if,” “when,” and “while” should be interpreted to mean “under the condition that” rather than imply an immediate temporal relationship or reaction. That is, these phrases, e.g., “when,” do not imply an immediate action in response to or during the occurrence of an action, but simply imply that if a condition is met then an action will occur, but without requiring a specific or immediate time constraint for the action to occur. The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may

include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as "at least one of A, B, or C," "one or more of A, B, or C," "at least one of A, B, and C," "one or more of A, B, and C," and "A, B, C, or any combination thereof" may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. Sets should be interpreted as a set of elements where the elements number one or more. Accordingly, for a set of X, X would include one or more elements. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words "module," "mechanism," "element," "device," and the like may not be a substitute for the word "means." As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase "means for."

(109) The following aspects are illustrative only and may be combined with other aspects or teachings described herein, without limitation.

(110) Aspect 1 is an apparatus for wireless communication at a UE, comprising: a memory; and at least one processor coupled to the memory and configured to: perform a spectrum sensing based on a set of resources or a configuration to acquire information about at least one of FD operation or CLI associated with one or more cells of a base station during initial access; and select a RO or a cell of the one or more cells based on the acquired information about the at least one of the FD operation or the CLI.

(111) Aspect 2 is the apparatus of any of aspects 1, wherein the set of resources comprises one or more time domain resources comprising one or more UL or FD symbols or slots associated with the spectrum sensing.

(112) Aspect 3 is the apparatus of any of aspects 2, wherein the one or more UL or FD symbols or slots associated with the spectrum sensing comprise all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a TDD configuration.

(113) Aspect 4 is the apparatus of any of aspects 1, wherein the set of resources comprises one or more frequency domain resources.

(114) Aspect 5 is the apparatus of any of aspects 1-4, wherein the one or more frequency domain resources comprise an initial UL BWP.

(115) Aspect 6 is the apparatus of any of aspects 1-5, wherein the one or more frequency domain resources comprise a dedicated BWP for the FD operation.

(116) Aspect 7 is the apparatus of any of aspects 1-6, wherein the one or more frequency domain resources comprise a collection of staggered RB groups.

(117) Aspect 8 is the apparatus of any of aspects 1-7, wherein the one or more frequency domain resources comprise at least one different frequency domain resource associated with at least one different time domain resource.

(118) Aspect 9 is the apparatus of any of aspects 1-8, wherein the set of resources comprises a time window prior to the RO.

(119) Aspect 10 is the apparatus of any of aspects 1-9, wherein the configuration comprises one or more conditions associated with a selection of the RO or the one or more cells.

(120) Aspect 11 is the apparatus of any of aspects 1-10, wherein the one or more conditions comprise one or more of: an average RSSI threshold, a highest RSSI threshold, or a number of resources associated with an RSSI above a threshold.

(121) Aspect 12 is the apparatus of any of aspects 1-11, wherein the set of resources overlaps with a set of resources configured for UL communication for a connected UE.

(122) Aspect 13 is the apparatus of any of aspects 1-12, wherein the cell of the one or more cells is a cell with a reduced chance of a high CLI.

(123) Aspect 14 is the apparatus of any of aspects 1-13, wherein the set of resources is selected based on the configuration, the configuration being a TDD configuration or a BWP configuration.
(124) Aspect 15 is the apparatus of any of aspects 1-14, wherein the information about the at least one of the FD operation or the CLI is acquired based on one or more FD resources.
(125) Aspect 16 is the apparatus of any of aspects 1-15, wherein the one or more FD resources comprise one or more TDD indications in RMSI.
(126) Aspect 17 is the apparatus of any of aspects 1-16, wherein the at least one processor is further configured to: measure at least one total RSSI associated with at least one resource of the set of resources.
(127) Aspect 18 is the apparatus of any of aspects 1-17, wherein the at least one processor is further configured to: measure at least one DL RSSI associated with one or more DL resources of the set of resources; and estimate at least one CLI RSSI based on the at least one DL RSSI and the total RSSI.
(128) Aspect 19 is the apparatus of any of aspects 1-17, wherein the at least one processor is further configured to: identify and distinguish an UL spectrum from a DL spectrum associated with the at least one resource.
(129) Aspect 20 is the apparatus of any of aspects 19, wherein the DL spectrum spans a wider BW or is associated with a more uniform PSD compared to the UL spectrum.
(130) Aspect 21 is the apparatus of any of aspects 1-20, wherein the at least one processor is further configured to: determine that the at least one resource comprises at least one UL resource and at least one DL resource.
(131) Aspect 22 is the apparatus of any of aspects 1-21, wherein the at least one processor is further configured to: determine that a cell associated with the at least one resource is an FD cell.
(132) Aspect 23 is the apparatus of any of aspects 1-22, further comprising a transceiver coupled to the at least one processor.
(133) Aspect 24 is an apparatus for wireless communication at a base station, comprising: a memory; and at least one processor coupled to the memory and configured to: transmit, to a UE, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about at least one of FD operation or CLI associated with one or more cells of the base station during initial access; and receive, from the UE, communication based on a selection of a RO or a cell of the one or more cells based on the information about the at least one of the FD operation or the CLI.
(134) Aspect 25 is the apparatus of aspect 24, wherein the set of resources comprises one or more time domain resources comprising one or more UL or FD symbols or slots associated with the spectrum sensing.
(135) Aspect 26 is the apparatus of any of aspects 24-25, wherein the one or more UL or FD symbols or slots associated with the spectrum sensing comprises all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a TDD configuration.
(136) Aspect 27 is the apparatus of any of aspects 24-26, wherein the set of resources comprises one or more frequency domain resources.
(137) Aspect 28 is the apparatus of any of aspects 24-27, further comprising a transceiver coupled to the at least one processor.
(138) Aspect 29 is a method of wireless communication for implementing any of aspects 1 to 23.
(139) Aspect 30 is an apparatus for wireless communication including means for implementing any of aspects 1 to 23.
(140) Aspect 31 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 1 to 23.
(141) Aspect 32 is a method of wireless communication for implementing any of aspects 24 to 28.
(142) Aspect 33 is an apparatus for wireless communication including means for implementing any of aspects 24 to 28.

(143) Aspect 34 is a computer-readable medium storing computer executable code, where the code when executed by a processor causes the processor to implement any of aspects 24 to 28.

Claims

1. An apparatus for wireless communication at a user equipment (UE), comprising: memory; and at least one processor coupled to the memory and configured to: perform, during initial access, a spectrum sensing based on a set of resources or a configuration to acquire information about full-duplex (FD) operation and cross link interference (CLI) associated with one or more cells of a base station, wherein to perform the spectrum sensing, the at least one processor is configured to: separately measure a first subset of downlink resources of the set of resources and a second subset of full-duplex resources of the set of resources, wherein the first subset of downlink resources is without corresponding uplink transmission of the UE, and wherein the second subset of full-duplex resources is associated with at least one uplink transmission of the UE; and identify a difference between a first measurement associated with the first subset of downlink resources and a second measurement associated with the second subset of full-duplex resources associated with the at least one uplink transmission of the UE to be corresponding to a CLI associated with at least other UE separate from the UE; and select a random access channel (RACH) occasion (RO) or a cell of the one or more cells based on the acquired information about the FD operation and the CLI.
2. The apparatus of claim 1, wherein the set of resources comprises one or more time domain resources comprising one or more uplink (UL) or FD symbols or slots associated with the spectrum sensing.
3. The apparatus of claim 2, wherein the one or more UL or FD symbols or slots associated with the spectrum sensing comprise all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a time division duplex (TDD) configuration.
4. The apparatus of claim 1, wherein the set of resources comprises one or more frequency domain resources.
5. The apparatus of claim 4, wherein the one or more frequency domain resources comprise an initial uplink (UL) bandwidth part (BWP).
6. The apparatus of claim 4, wherein the one or more frequency domain resources comprise a dedicated bandwidth part (BWP) for the FD operation.
7. The apparatus of claim 4, wherein the one or more frequency domain resources comprise a collection of staggered resource block (RB) groups.
8. The apparatus of claim 4, wherein the one or more frequency domain resources comprise at least one different frequency domain resource associated with at least one different time domain resource.
9. The apparatus of claim 1, wherein the set of resources comprises a time window prior to the RO.
10. The apparatus of claim 1, wherein the configuration comprises one or more conditions associated with a selection of the RO or the one or more cells.
11. The apparatus of claim 10, wherein the one or more conditions comprise one or more of: an average received signal strength indicator (RSSI) threshold, a highest RSSI threshold, or a number of resources associated with an RSSI above a threshold.
12. The apparatus of claim 10, wherein the set of resources overlaps with a set of resources configured for UL communication for a connected UE.
13. The apparatus of claim 1, wherein the cell of the one or more cells is a cell with a reduced chance of a high CLI.
14. The apparatus of claim 1, wherein the set of resources is selected based on the configuration, the configuration being a time division duplex (TDD) configuration or a bandwidth part (BWP) configuration.
15. The apparatus of claim 1, wherein the information about the FD operation and the CLI is

acquired based on one or more FD resources.
16. The apparatus of claim 1, wherein the one or more FD resources comprise one or more time division duplex (TDD) indications in remaining minimum system information (RMSI).
17. The apparatus of claim 1, wherein the at least one processor is further configured to: measure at least one total received signal strength indicator (RSSI) associated with at least one resource of the set of resources.
18. The apparatus of claim 17, wherein the at least one processor is further configured to: measure at least one DL RSSI associated with one or more DL resources of the set of resources; and estimate at least one CLI RSSI based on the at least one DL RSSI and the total RSSI.
19. The apparatus of claim 17, wherein the at least one processor is further configured to: identify and distinguish an UL spectrum from a DL spectrum associated with the at least one resource.
20. The apparatus of claim 19, wherein the DL spectrum spans a wider bandwidth (BW) or is associated with a more uniform power spectrum density (PSD) compared to the UL spectrum.
21. The apparatus of claim 19, wherein the at least one processor is further configured to: determine that the at least one resource comprises at least one UL resource and at least one DL resource.
22. The apparatus of claim 21, wherein the at least one processor is further configured to: determine that a cell associated with the at least one resource is an FD cell.
23. The apparatus of claim 1, further comprising a transceiver coupled to the at least one processor, wherein the at least one processor is further configured to: receive, from the base station, system information indicating the set of resources or the configuration for the spectrum sensing.
24. An apparatus for wireless communication at a base station, comprising: memory; and at least one processor coupled to the memory and configured to: transmit, to a user equipment (UE) during initial access, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about full-duplex (FD) operation and cross link interference (CLI) associated with one or more cells of the base station; configure, in the configuration for the spectrum sensing, a first subset of downlink resources of the set of resources and a second subset of full-duplex resources of the set of resources, wherein the first subset of downlink resources is without corresponding uplink transmission of the UE, and wherein the second subset of full-duplex resources is associated with at least one uplink transmission of the UE; and receive, from the UE, communication based on a selection of a random access channel (RACH) occasion (RO) or a cell of the one or more cells based on the information about the FD operation and the CLI.
25. The apparatus of claim 24, wherein the set of resources comprises one or more time domain resources comprising one or more uplink (UL) or FD symbols or slots associated with the spectrum sensing.
26. The apparatus of claim 25, wherein the one or more UL or FD symbols or slots associated with the spectrum sensing comprises all UL symbols or slots, all flexible symbols or slots, or all FD symbols or slots associated with a time division duplex (TDD) configuration.
27. The apparatus of claim 24, wherein the set of resources comprises one or more frequency domain resources.
28. The apparatus of claim 24, further comprising a transceiver coupled to the at least one processor.
29. A method for wireless communication at a user equipment (UE), comprising: performing, during initial access, a spectrum sensing based on a set of resources or a configuration to acquire information about full-duplex (FD) operation and cross link interference (CLI) associated with one or more cells of a base station, wherein performing the spectrum sensing comprises: separately measuring a first subset of downlink resources of the set of resources and a second subset of full-duplex resources of the set of resources, wherein the first subset of downlink resources is without corresponding uplink transmission of the UE, and wherein the second subset of full-duplex

resources is associated with at least one uplink transmission of the UE; and identifying a difference between a first measurement associated with the first subset of downlink resources and a second measurement associated with the second subset of full-duplex resources associated with the at least one uplink transmission of the UE to be corresponding to a CLI associated with at least other UE separate from the UE; and selecting a random access channel (RACH) occasion (RO) or a cell of the one or more cells based on the acquired information about the FD operation and the CLI.

30. A method for wireless communication at a base station, comprising: transmitting, to a user equipment (UE) during initial access, system information indicating a set of resources or a configuration for spectrum sensing, the spectrum sensing being associated with acquiring information about full-duplex (FD) operation and cross link interference (CLI) associated with one or more cells of the base station; configuring, in the configuration for the spectrum sensing, a first subset of downlink resources of the set of resources and a second subset of full-duplex resources of the set of resources, wherein the first subset of downlink resources is without corresponding uplink transmission of the UE, and wherein the second subset of full-duplex resources is associated with at least one uplink transmission of the UE; and receiving, from the UE, communication based on a selection of a random access channel (RACH) occasion (RO) or a cell of the one or more cells based on the information about the FD operation and the CLI.